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# **ANALYTICAL REVIEW OF 5G NR CHANNEL CODING TECHNIQUES LDPC AND POLAR CODES**

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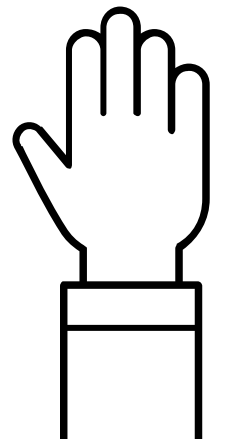
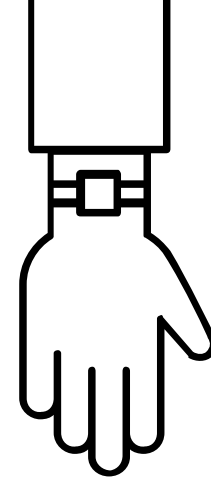
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# Presentation Outline

- Introduction
- Motivation
- Literature Survey & Objective
- Methodology
- Result Analysis
- Conclusion
- References



# Introduction

5G has mainly three service scenarios

| Use Cases                                        | Requirements                                                                                                    |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Ultra-Reliable Low-Latency Communication (uRLLC) | To support low-latency transmissions of small blocks with very high reliability from a limited set of terminals |
| Enhanced Mobile Broadband (eMBB)                 | To support stable connections with very high peak data rates, as well as moderate rates for cell-edge users     |
| Massive Machine Type Communication (mMTC)        | To support massive number of Internet of Things (IoT) device                                                    |

- The 5G New Radio (NR) air interface is a major transition to fulfil dynamic and flexible service requirements.
- The network needs to handle extremely high data rates with very high reliability, and enhanced quality of service [1].
- New channel coding techniques need to be analyzed to reduce redundancy and complexity in the wireless communication networks.

# Motivation

- The channel coding schemes are expected to provide flexibility for block length and code rates.
- Channel coding enhances the reliability and to be done in an optimized way to effectively enhance performance without unnecessary overhead bits [2].
- The key enabling channel codes for eMBB, mMTC, URLLC are LDPC codes and Polar codes.
- The majority of available literature does not take into account the specific modulation orders, decoding algorithms, and code rate impact on the designed channel codes for 5G; the research community would benefit from considering them within error-correction performance evaluations and BER analysis.

**Objective**: The objective of this paper is to present an analytical framework is discussed to understand the numerology required for LDPC and polar coded systems. 5G NR is the imminent evolution of next generation mobile technology to enhance spectral efficiency, signaling efficiency, data rate, and connection density.

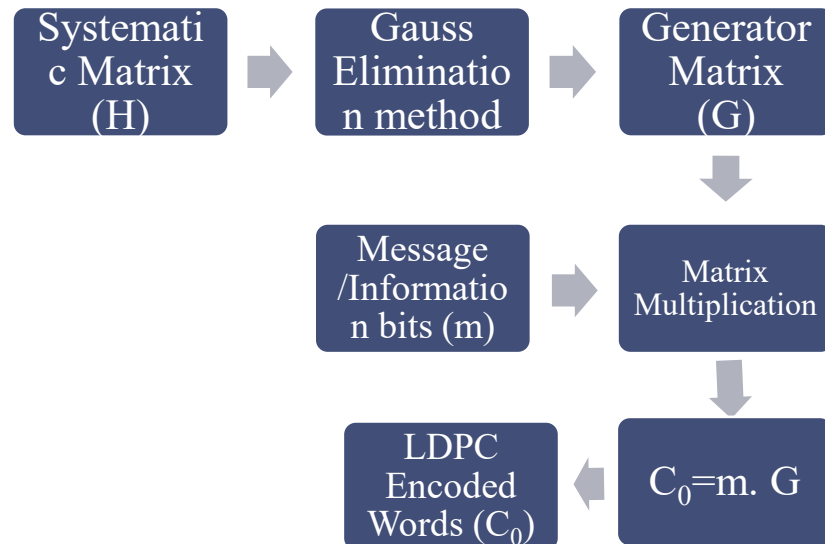
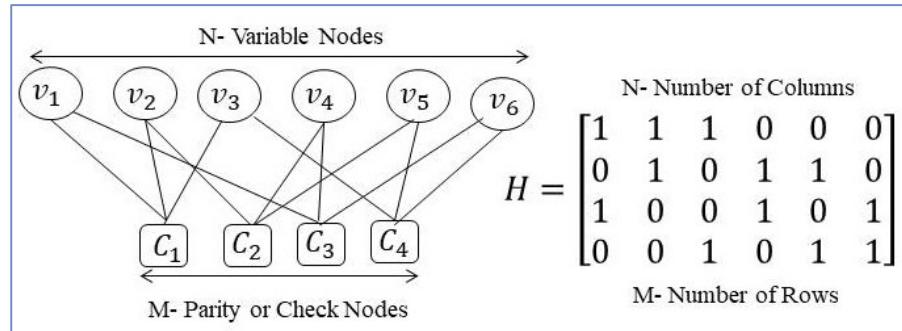
## Literature Survey

- LDPC codes are mainly applied for the data channel and targeted to provide high throughput and flexible code rate. The polar control codes are designed particularly for the control channel with short block length to resolve the latency issue needed in URLLC use case [7].

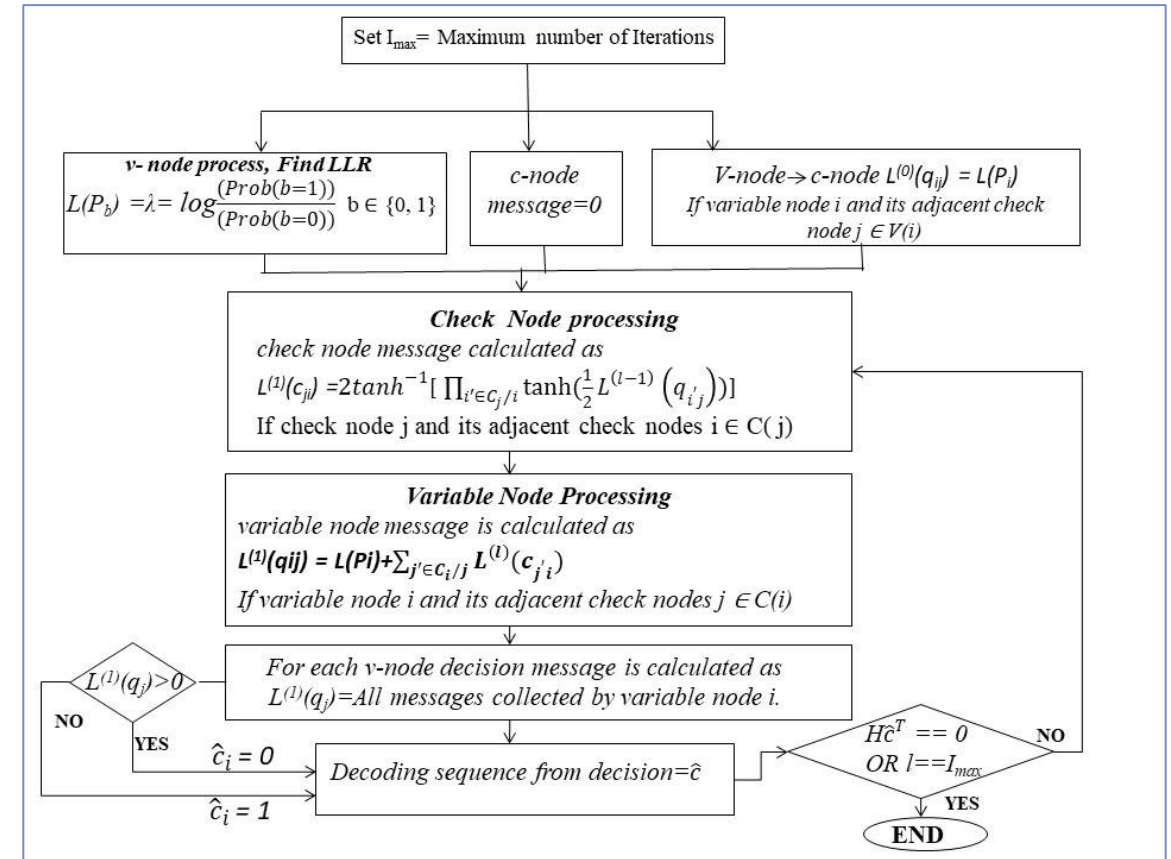
| LDPC Codes          |                                                                                        |                                                                                            |            |
|---------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------|
| Details             | Description                                                                            | Advantages                                                                                 | References |
| Encoding Technique  | LDPC codes are described via matrices or represented graphically (Tanner graphs).      | Improved area throughput efficiency and error performance                                  | [8-10]     |
|                     | Quasi-cyclic (QC) LDPC coding based on the cyclic shift                                | This code supports variable code lengths                                                   | [11, 12]   |
| Decoding Algorithms | Belief propagation algorithm                                                           | Reduced decoding complexity                                                                | [8, 13]    |
|                     | Weighted bit flipping algorithm (WBFA)                                                 | Improved decoding latency                                                                  | [14, 15]   |
|                     | Sum-product algorithm                                                                  | Design flexibility in selecting the appropriate design point for high-speed applications   | [16]       |
| Polar Codes         |                                                                                        |                                                                                            |            |
| Encoding Technique  | Polar coding is done recursively via Kronecker products termed as channel polarization | It achieves the symmetric (Shannon) capacity of a binary input discrete memoryless channel | [17]       |
| Decoding Algorithms | Successive cancellation (SC)                                                           | Based on depth-first binary tree search                                                    | [17]       |
|                     | SC-List decoding                                                                       | SCL improves the error-correction performance, complexity increases as well.               | [18, 19]   |

# LDPC Code Design

- LDPC codes use a sparse (minimum number of ones and majority is zeros in the matrix) parity check matrix (PCM) for encoding. The encoding algorithm is applied to transform the input K bit segment into an output block of length N.
- The visual representation of PCM is done by a Tanner graph [22].



## LDPC Encoding Algorithms



## Belief Propagation Decoding Algorithms

# Methodology

## Polar Code Design

### Polar Coding Algorithms

Input vector is generated as per equation:

$$u_i = \begin{cases} 0, & \text{if } i \in F \\ \text{Unchanced}, & \text{if } i \in I \end{cases} \quad (1)$$

Code word  $x = [x_0, x_1, \dots, x_{N-1}]$  is then calculated as

$$x = u \cdot G_N \quad (2)$$

$$G_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad (3)$$

$$G_N = G_2^{\otimes n} \quad (4)$$

$$G_N = G_{N/2}^{\otimes_2} = \begin{pmatrix} G_{N/2} & 0 \\ G_{N/2} & G_{N/2} \end{pmatrix} \quad (5)$$

Let's model  $W$  to the MBIOS channel implement the transform that polarizes the output.

$$y = x \cdot W \quad (6)$$

with  $W_N^{(i)}$  are synthesized binary channel where  $i = 1, 2, \dots, N$ .

### Successive Cancellation Decoding Algorithm

**Initialization:** *Coder length*  $N$ ;  
Encoder input bit  $u_i$ ;  
Synthesized binary channels  $W_N^{(i)}$   
for  $0 \leq i < N$

*Channel Transition*  
probability  $= W_N^{(i)}(y_o^{N-1}, u_o^{i-1} | u_i)$

$\hat{u}_i = \text{Estimate of Source block}$   
from received block  $y_1^N$   
Evaluate  $h_i(y_1^N, u_1^{i-1})$

If  $\frac{W_N^{(i)}(y_1^{N-1}, \hat{u}_1^{i-1} | 0)}{W_N^{(i)}(y_1^{N-1}, \hat{u}_1^{i-1} | 1)} \geq 1$  **True**  $h_i(y_1^N, u_1^{i-1}) = 0$

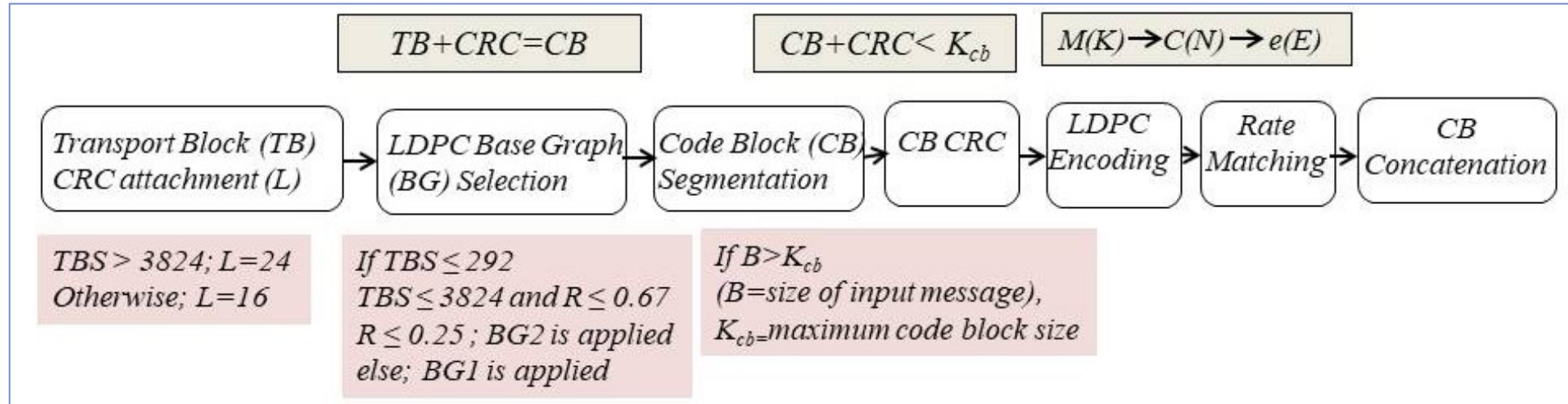
**False**

$$h_i(y_1^N, u_1^{i-1}) = 1$$

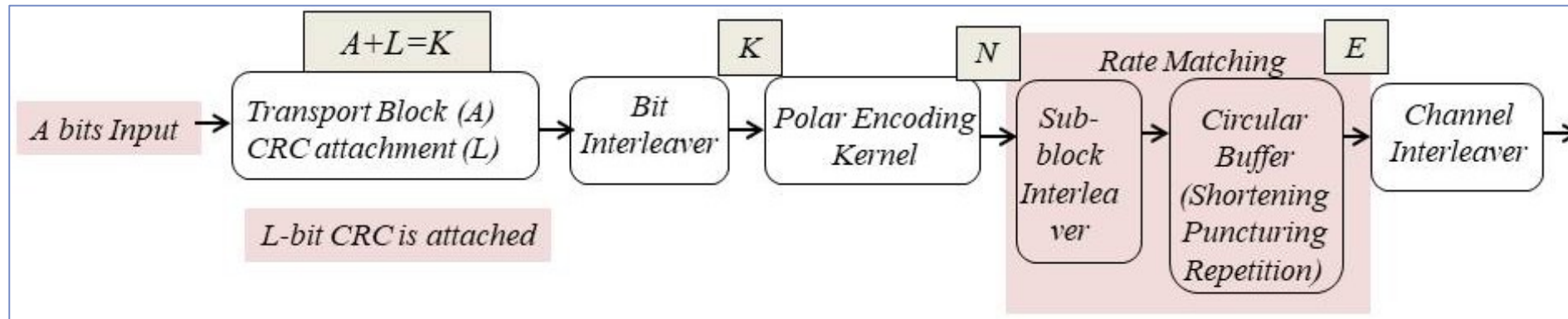
$$\hat{u}_i = \begin{cases} h_i(y_1^N, u_1^{i-1}), & i \in I \\ u_i, & i \in F \end{cases}$$

# System Design For simulation

## LDPC Coding



## Polar Coding

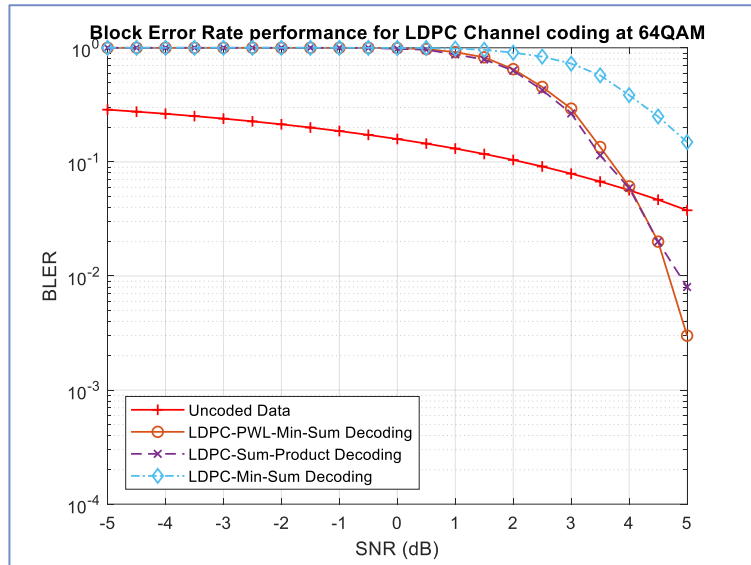


# Simulation and Results

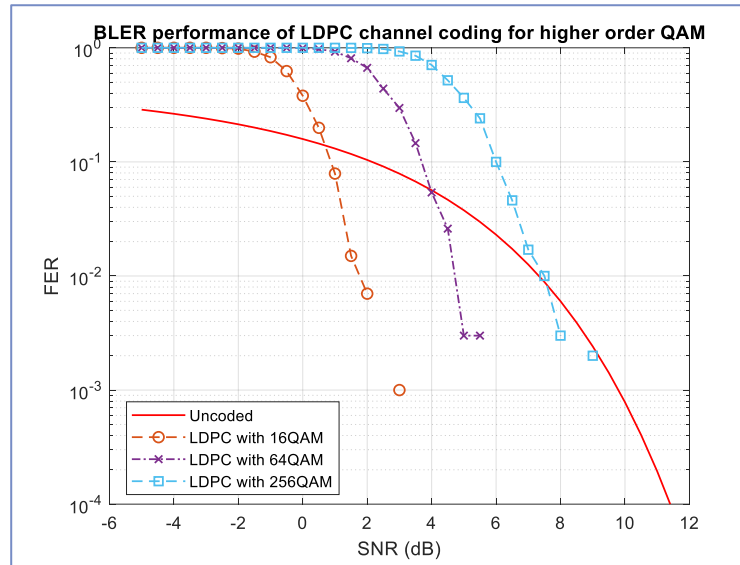
| Parameters              | Specifications            |                 |                                    |                                  |         |                   |
|-------------------------|---------------------------|-----------------|------------------------------------|----------------------------------|---------|-------------------|
|                         | AWGN                      |                 |                                    |                                  |         |                   |
| Channel                 |                           |                 |                                    |                                  |         |                   |
| Modulation              | QAM (Order=16, 64, 256)   |                 |                                    |                                  |         |                   |
| Input Block Size        | 188                       |                 |                                    |                                  |         |                   |
| CRC Length              | 16                        |                 |                                    |                                  |         |                   |
| Number of Frames        | 1000                      |                 |                                    |                                  |         |                   |
| HARQ Type               |                           |                 |                                    |                                  |         |                   |
|                         | Chase Combining (CC)      |                 |                                    |                                  |         |                   |
| Redundancy Version (RV) |                           |                 |                                    |                                  |         |                   |
|                         | RV0                       |                 |                                    |                                  |         |                   |
| Channel Coding          | LDPC (5G NR based coding) |                 |                                    | Polar (PC) (Unsystematic coding) |         |                   |
| Decoding Algorithm      | Layered Sum-Product       | Layered Min-Sum | Layered (Piecewise Linear Min-Sum) | Successive Cancellation (SC)     | SC List | CRC List-SC aided |
| Code Rate               | 0.256                     |                 |                                    | 1/6                              |         |                   |
| Specs applicable        | Base Graph 2              |                 |                                    | List Size=32                     |         |                   |

# Simulation and Results

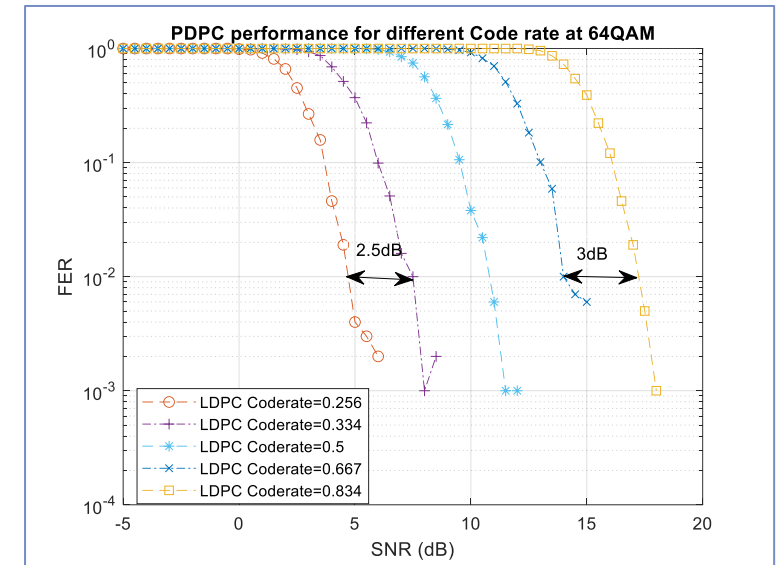
## LDPC Coding: BLER Vs SNR



(a)



(b)

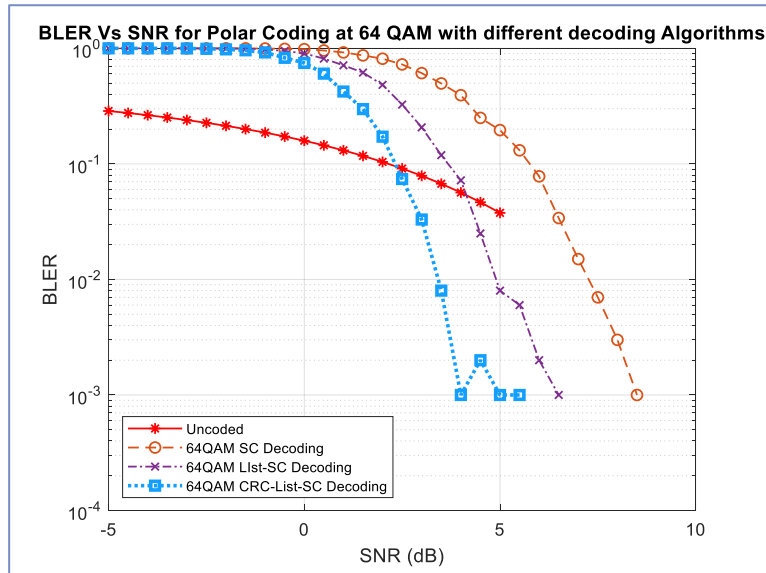


(c)

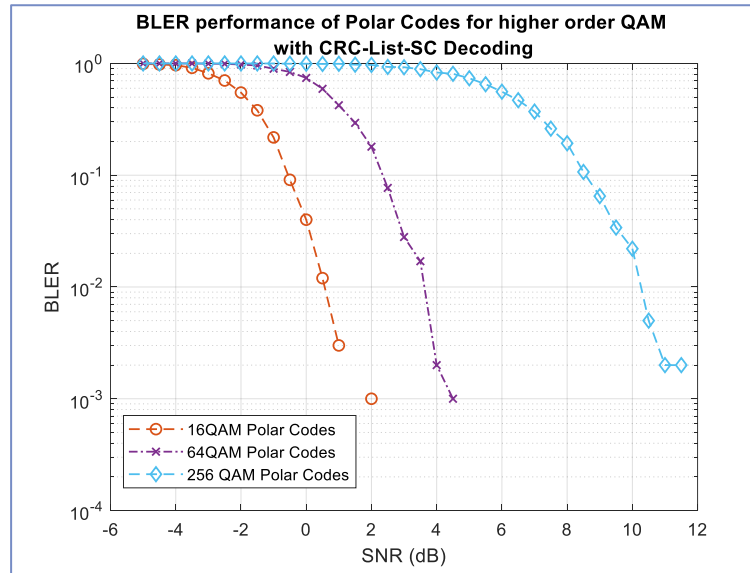
**(a)** BLER Vs SNR for LDPC Coding for different decoding algorithms **(b)** Comparative Plot for LDPC with PWL Min-Sum decoding for higher order QAM **(c)** LDPC with variable code rate PWL Min-Sum Decoding at 64 QAM

# Simulation and Results

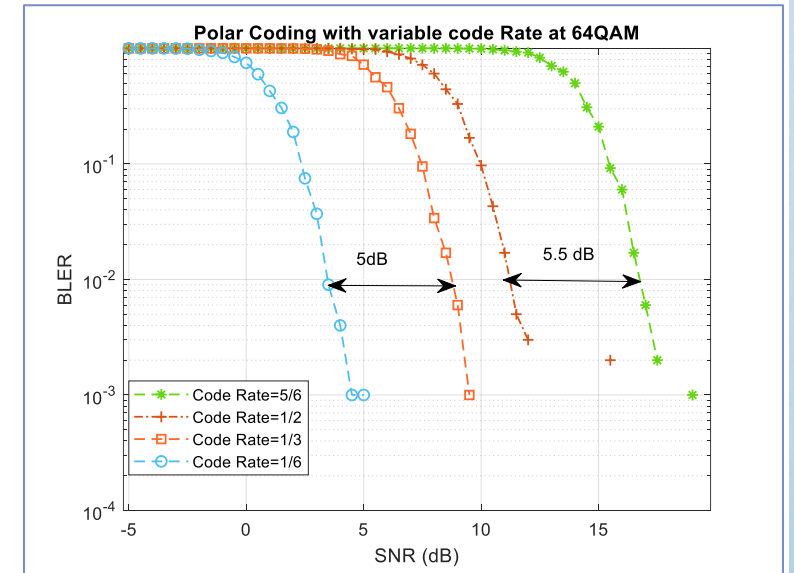
## Polar Coding: BLER Vs SNR



(a)



(b)

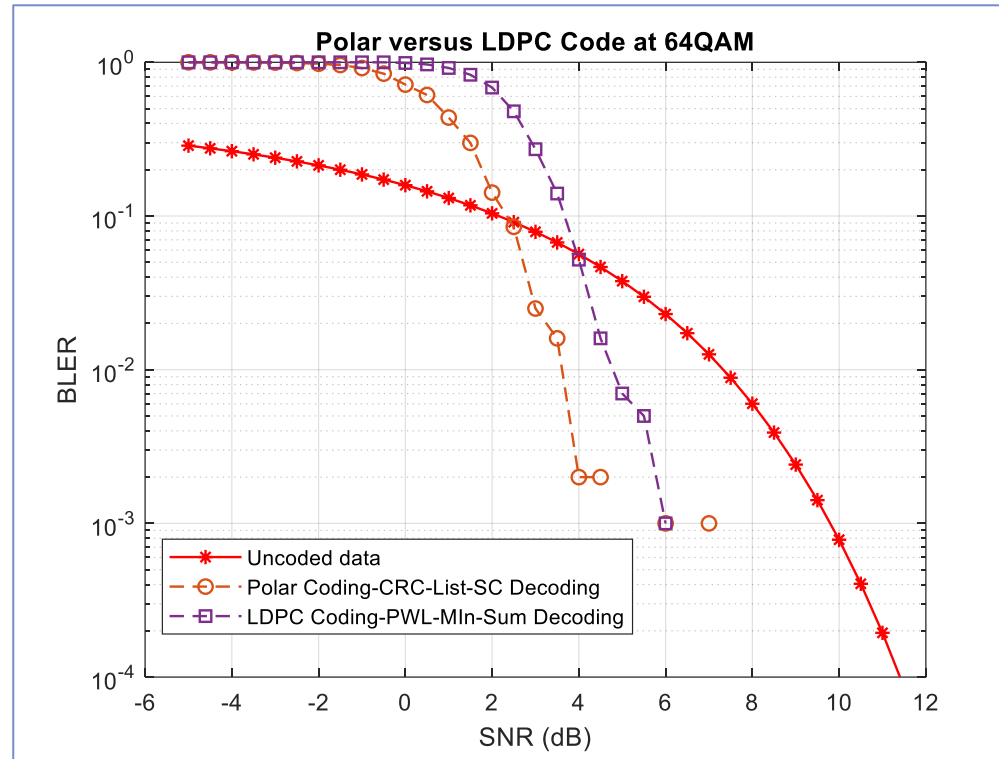


(c)

(a) (b) Polar Coding for different decoding algorithms and QAM Order (c) 5G NR Polar channel coding with variable Code Rate

# Simulation and Results

## Polar Coding Vs LDPC Coding



## Polar channel coding versus LDPC Coding at 64QAM

## Conclusion

- In this paper, performance of LDPC and Polar codes are observed and analysed for scenarios of URLLC and mMTC for short block transmission.
- Several performance aspects of candidate coding schemes for 5G NR are analyzed.
- We have presented the comparative analysis of the error rate for the LDPC and Polar codes.
- The codes are simulated with different decoding techniques, code rates and QAM modulation order.
- It is found that LDPC codes have high computational complexity as compared to the polar codes.
- Polar codes provide flexibility with better error rate performance in any code length and code rate.

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